Operational KPI Definitions, Alert Thresholds and Escalation Matrix

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1. Purpose

This document defines the key performance indicators used on the CareOps AI dashboard, explains how each one is calculated, and lists the alert thresholds and escalation steps used at CareOps General Hospital. Formula definitions follow standard hospital statistics practice. Thresholds are synthetic internal policy values.

2. Bed and inpatient KPIs

Bed occupancy rate is the number of occupied beds divided by the number of staffed available beds, multiplied by 100. Over a period it can also be calculated as occupied bed days divided by available bed days, multiplied by 100. Available beds are beds that are vacant, clean and staffed. Average length of stay is the total number of inpatient days divided by the number of discharges in the same period. Bed turnover rate is the number of discharges divided by the number of available beds in a period, and it shows how many patients each bed served. Bed turnover interval is the average time a bed stays empty between one patient leaving and the next patient arriving. A long turnover interval suggests cleaning or allocation delays. Discharge before noon rate is the percentage of discharges completed before 12:00, and the target is 30 percent.

3. Emergency Department KPIs

Door-to-triage time is the time from arrival to triage assessment, with a target of 5 minutes. Door-to-doctor time, also called waiting time, is the time from arrival to first contact with a doctor. ED length of stay is the time from arrival to the moment the patient leaves the ED, either by discharge, admission or transfer. The target is that 90 percent of ED patients are discharged, admitted or transferred within 4 hours. ED boarding is the number of admitted patients who are still in the ED after the decision to admit has been made, and boarding for more than 2 hours is reported as delayed. Left without being seen rate is the percentage of registered patients who leave before seeing a doctor, with a target below 2 percent. ED occupancy is occupied ED beds divided by the 40 ED beds.

4. Staffing, equipment and outpatient KPIs

Staff utilization is the hours worked divided by the hours scheduled, multiplied by 100. Nurse-to-patient ratio is the number of patients assigned to each nurse on a shift. Staff availability is the number of staff present divided by the number required by the roster, expressed as a percentage. Equipment utilization is the hours the equipment was in use divided by the hours it was available, multiplied by 100. Equipment downtime is the share of time equipment is out of service. OPD no-show rate is missed appointments divided by booked appointments, multiplied by 100. OT utilization is the scheduled theatre time actually used divided by the available theatre time.

5. Alert levels

The dashboard uses four alert colours. Green means normal operations. Yellow means watch and requires acknowledgement within 30 minutes. Orange means high pressure and requires acknowledgement within 15 minutes. Red means critical and requires acknowledgement within 5 minutes. When several indicators trigger different colours, the highest colour applies to the department or hospital.

6. Threshold values

Hospital-wide bed occupancy is Green below 75 percent, Yellow from 75 to 84 percent, Orange from 85 to 89 percent, and Red at 90 percent or more. ED occupancy is Green below 70 percent, Yellow from 70 to 79 percent, Orange from 80 to 89 percent, and Red at 90 percent or more. ICU occupancy is Yellow from 80 to 84 percent, Orange from 85 to 94 percent and Red at 95 percent or more. ED average waiting time is Green below 30 minutes, Yellow from 30 to 44 minutes, Orange from 45 to 59 minutes and Red at 60 minutes or more. ED arrivals are normal at 20 to 40 per hour, Yellow above 40 and up to 60 per hour, Orange above 60 and up to 100 per hour, and Red above 100 per hour. Staff availability is Green at 95 percent or more of the roster requirement, Yellow from 85 to 94 percent, Orange from 70 to 84 percent, and Red below 70 percent. Staff availability labeled Low in dashboard summaries means Orange or worse. Critical equipment availability for ventilators and defibrillators is Yellow when fewer than 20 percent are free, and Red when fewer than 10 percent are free.

7. Anomaly interpretation

An anomaly is a value that differs strongly from the usual pattern for the same department, day of week and hour. For example, if a department usually receives 20 to 40 patients per hour and suddenly receives 100, the anomaly detector flags the change even before a fixed threshold is crossed. An anomaly alert means the pattern is unusual and should be investigated. It does not prove that an emergency is happening. Common reasons for an arrival spike are a road traffic accident, a mass gathering, an outbreak, a local disease surge, or weather. Common reasons for an unusual rise in equipment usage are a patient surge, a device that is left running, or a sensor error.

8. Escalation matrix

A Yellow alert is handled by the Charge Nurse, who informs the Bed Manager. An Orange alert is handled by the Duty Manager, who informs the Medical Superintendent. A Red alert is handled by the Medical Superintendent, who informs the Hospital Administrator, and the matter is reviewed every 2 hours until the alert returns to Orange or below. Equipment alerts go to the Biomedical Engineer on duty. Infection-related alerts go to the Infection Control Officer. An alert is closed only when the indicator stays below its threshold for 2 consecutive hours and the owner records a closing note.